

1 Background and Challenges

- ▶ **The single-report paradigm misses pathologies:** existing 3D CT systems decode one report per volume by greedy or beam search, so many of the 18 pathologies are never mentioned. Under clinical-efficacy (CE) F1 this shows as high precision with low recall, **0.44** against **0.18**.
- ▶ **Generalization makes it harder:** the same recall limitation grows once evaluation moves to different datasets.

2 Contributions

- ▶ **A deep-ensemble architecture.** We propose MDEF, a five-stage pipeline that aggregates multiple generated reports at the 18-pathology binary-vector level by logical-OR, so any pathology mentioned in any sampled report reaches the final report. To the best of our knowledge, MDEF is the first such pipeline for 3D CT report generation, and it raises micro-averaged recall by **+70%** to **+97%** over MARCH.
- ▶ **A generalization evaluation.** We train MDEF on RadGenome and evaluate it under one fixed configuration on three different held-out datasets. Our MDEF reaches the highest CE F1 on all three, beating MARCH by **+0.079**, **+0.045** and **+0.008**.

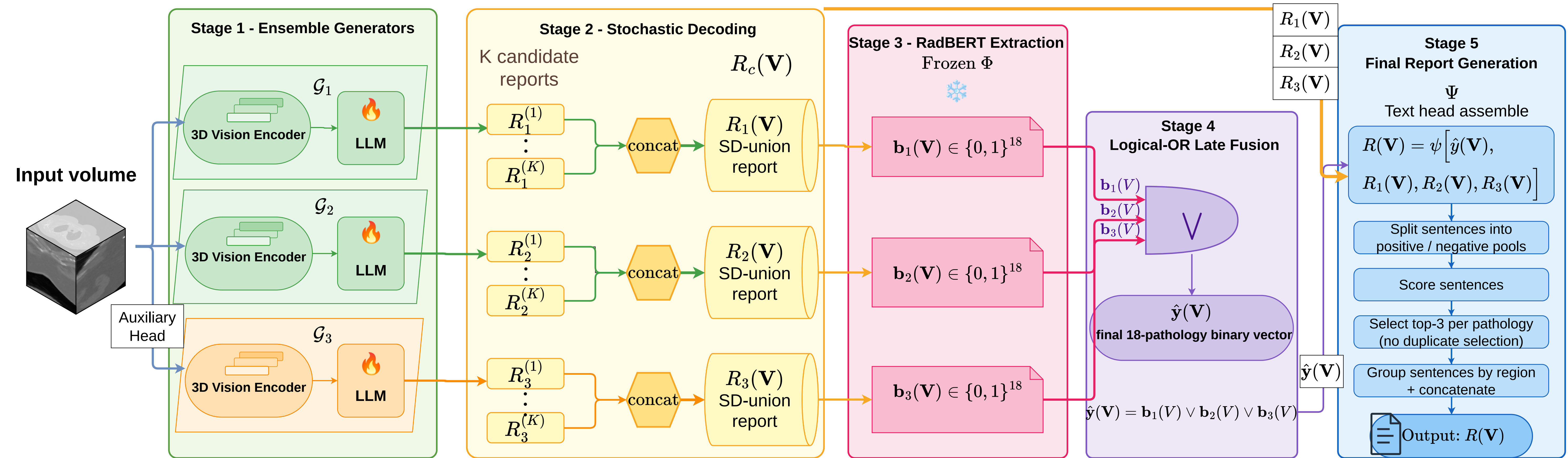
4 Experimental Setup

- ▶ **Trained on RadGenome (RG) only**, with one fixed configuration applied unchanged to every held-out dataset.
- ▶ **Three different held-out datasets:** in-distribution RG, cross-dataset CT-RATE (CTR) and cross-institution INSPECT (INS).
- ▶ **Metrics:** clinical-efficacy (CE) F1 over the 18 RadBERT pathologies, and the NLG metrics BLEU, METEOR and ROUGE-L.
- ▶ **Baselines:** Reg2RG, CT-GRAPH and MARCH, the strongest single-report generators.
- ▶ **Configuration:** our MDEF and all three baselines share a 3D vision encoder with Llama-2-7B as the language model.

7 Conclusion

- ▶ **Method:** MDEF turns the single-report paradigm into a multi-report one and fuses reports at the 18-pathology vector level.
- ▶ **Performance:** our MDEF reaches the highest CE F1 on all three different held-out datasets under one fixed configuration.
- ▶ **Recall gain:** the improvement is driven by a per-pathology recall gain over every single-report baseline.
- ▶ **Future work:** addressing the precision cost of the logical-OR aggregator, and broadening the evaluation to more chest-CT datasets.

3 Overall Architecture



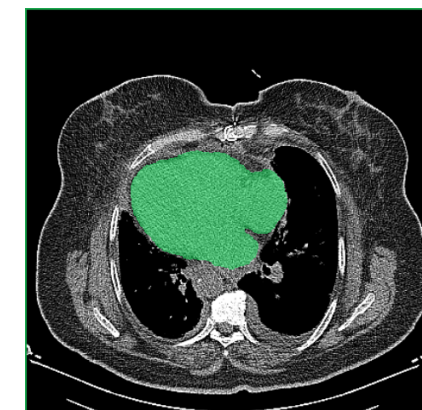
- ▶ **Stage 1 and Stage 2:** three generators each draw $K=10$ Stochastic Decoding candidates that are concatenated into one SD-union report.
- ▶ **Stage 3:** RadBERT maps every SD-union report to an 18-pathology binary vector.
- ▶ **Stage 4:** a per-pathology logical-OR fuses the three vectors, so the ensemble misses a pathology only when every generator misses it.
- ▶ **Stage 5:** the deterministic text head scores candidate sentences, selects the highest scoring ones per pathology and groups them by anatomical region.

5 Empirical Results

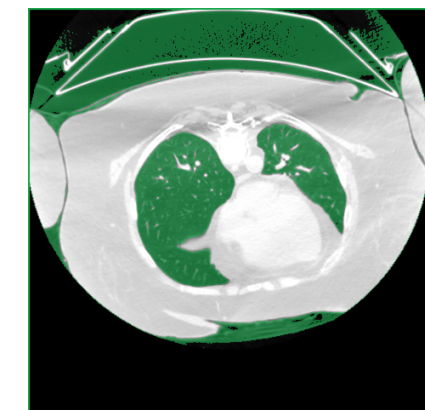
CE F1	RG	CTR	INS
Reg2RG	.2564	.1130	.1329
CT-GRAPH	.2498	.1032	.1256
MARCH	.3214	.2934	.1738
Our MDEF	.3999	.3381	.1813

- ▶ **Highest CE F1 on all three datasets**, beating MARCH by **+0.079**, **+0.045** and **+0.008**.
- ▶ **The lift is driven by recall**, which rises by **+70%** to **+97%** over MARCH.
- ▶ **Per-pathology recall:** higher than MARCH on **18 of 18** pathologies on RG, **14 of 18** on CTR and **13 of 18** on INS.
- ▶ **NLG metrics:** best on **9 of the 12** BLEU, METEOR and ROUGE-L scores on CTR and INS.
- ▶ **Efficiency:** **13%** lower wall time than MARCH.

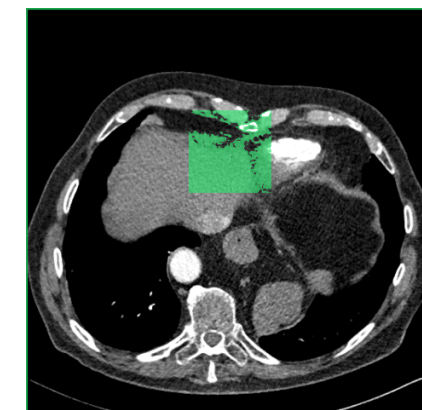
6 Qualitative Evidence



RG / pericardial effusion
GT. "Pericardial effusion at the apex, ~12 mm. No pneumothorax."
Reg2RG. "Mediastinal contours unremarkable."
CT-GRAPH. "Heart size within normal limits. Lungs clear."
MARCH. "No pneumothorax. Mediastinum and hila unremarkable."
Our MDEF. "12 mm pericardial effusion observed at the apex."



CTR / atelectasis
GT. "Atelectatic changes in the lower lobes. No pleural effusion."
Reg2RG. "Mediastinum and hila unremarkable."
CT-GRAPH. "Lung parenchyma is otherwise clear."
MARCH. "No pleural effusion. Heart size within normal limits."
Our MDEF. "Atelectasis observed in the lower-lobe posterobasal segment."



INS / hiatal hernia
GT. "Large hiatal hernia, intrathoracic stomach. No pulmonary embolus."
Reg2RG. "No pulmonary embolus. Heart size normal."
CT-GRAPH. "Lungs clear, no acute thoracic pathology."
MARCH. "No pleural effusion. No pneumothorax."
Our MDEF. "Hiatal hernia is observed."